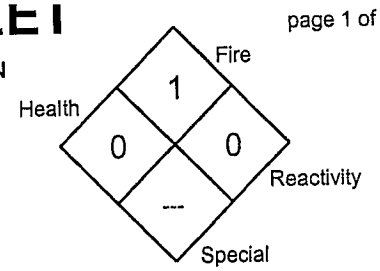
MATERIAL SAFETY DATA SHEET

GW International MSDS #	28709
MSDS Revision/Issue Date:	06/14/00
Supercedes Revision Date:	06/30/97

NFPA 704 DESIGNATION
HAZARD RATING

4=Extreme
3=High
2=Moderate
1=Slight
0=Insignificant



1. CHEMICAL PRODUCT IDENTIFICATION & COMPANY IDENTIFICATION

PRODUCT IDENTIFIER: **PROPYLENE GLYCOL, All Grades**

GENERAL USE: Uses: a non-toxic antifreeze in breweries and dairies; a substitute for Ethylene Glycol and Glycerol; in synthetic resin manufacturing; an inhibitor of fermentation and mold growth; & an emulsifier in food.

PRODUCT DESCRIPTION: A clear, colorless, organic chemical of the glycol family. Synonyms include: 1,2-Dihydroxypropane, Methylethylene Glycol, Methyl Glycol, Monopropylene Glycol, 1,2-Propanediol, 1,2-Propylene Glycol and Trimethyl Glycol.

INFORMATION PROVIDED BY:
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For MSDS call: **PHONE: 503-228-2600 or 800-547-1400**

EMERGENCY PHONE NUMBERS

GW INTERNATIONAL:	**800-497-7455**
CHEMTREC:	**800-424-9300**
CANUTEC:	**613-996-6666**

2. COMPOSITION & INFORMATION ON INGREDIENTS

COMPONENT	CAS #	OSHA HAZARD	WT %	ACGIH TLV(TWA)	ACGIH STEL	OSHA PEL(TWA)	OSHA STEL
Propylene Glycol	000057-55-6	Mild Eye, Skin Irritant	100	None	None	None	None

NDA = No Data Available N/A = Not Applicable

3. HAZARDS IDENTIFICATION

EMERGENCY OVERVIEW: A clear, colorless liquid having a very slight, mild odor. Mists and liquid are mildly irritating to the eyes and skin. Ingestion may cause a mild gastrointestinal irritation and/or discomfort.

POTENTIAL HEALTH EFFECTS

INHALATION: No significant adverse effects have been reported from inhalation of this material.

EYE CONTACT: Exposure to liquid or mists may cause temporary irritation if not washed out and is left in the eye for an extended period of time. Symptoms of exposure may include tearing, redness and a possible stinging sensation. Corneal damage or visual impairment is not expected to result from contact with this material.

SKIN CONTACT: Exposure to liquid or mists may cause a mild irritation. Direct contact with liquid or mists, for an extended period of time, trapped in clothing, shoes, etc., may cause moderate to severe irritation. This irritant effect is not expected to result in any permanent damage. There is no published data indicating this material is absorbed through the skin.

INGESTION: Ingestion may cause mild irritation to the entire gastrointestinal tract, including the stomach and intestines. Symptoms of exposure may include nausea, vomiting, lethargy and/or diarrhea.

CHRONIC: The chronic exposure effects are expected to be the same as for acute exposure.

4. FIRST AID MEASURES

INHALATION: If inhaled, and breathing is difficult, immediately move to fresh air. If symptoms persist, call a physician.

EYE CONTACT: In case of contact, immediately flush eyes with plenty of clean running water for at least 15 minutes, lifting the upper and lower lids occasionally. Remove contact lenses, if worn. Get medical attention if irritation persists.

SKIN CONTACT: In case of contact, flush skin with plenty of clean running water. Remove contaminated clothing and shoes, and wash before reuse. If irritation occurs and persists, get medical attention.

INGESTION: If swallowed, give plenty of water to drink. Call a physician. DO NOT induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person.

NOTE TO PHYSICIANS: Treat exposure symptomatically.

5. FIRE FIGHTING MEASURES

Flashpoint and Method: 101.1 °C (214 °F) Tagliabue Closed Cup Tester (ASTM D 56-79)

Flammable Limits (in air, % by volume) **Lower:** 2.6% **Upper:** 12.5%

Autoignition Temperature: 371 °C (700 °F)

GENERAL HAZARD: This material is not a combustible liquid per OSHA or WHMIS criteria, but it can be ignited at elevated temperatures and will burn. The Uniform Fire Code physical hazard classification for this material is: **Combustible Liquid, Class III-B.** It may produce hazardous fumes or hazardous decomposition products. Excessively heated containers may rupture violently in a fire.

FIRE FIGHTING INSTRUCTIONS: **EXTINGUISHING MEDIA:** Water fog, foam, CO_2 or dry chemicals.

Use water spray or fog to cool containers exposed to fire. Do not direct solid stream of water or foam into hot, burning pools; this may cause frothing and increase the fire's intensity.

FIRE FIGHTING EQUIPMENT: Fire fighters should wear full protective equipment, including self-contained breathing apparatus.

HAZARDOUS COMBUSTION PRODUCTS: When heated to decomposition, it emits toxic carbon monoxide and carbon dioxide plus possible dense smoke.

6. ACCIDENTAL RELEASE MEASURES

LAND SPILL: Wearing recommended protective equipment and clothing, dike the spill and pick up the bulk of liquid using pumps or a vacuum truck for potential recovery and return to the appropriate container. Absorb the remaining liquid using sand or a commercial absorbent; dispose as non-hazardous solid waste. Flush the spill area with water; collect the rinsates for disposal or sewer, as appropriate.

WATER SPILL: Wear recommended protective equipment and clothing if contact with hazardous material can occur. Stop or divert water flow. Dike contaminated water and remove for disposal and/or treatment. As appropriate, notify all downstream users of possible contamination.

7. HANDLING AND STORAGE

STORAGE TEMPERATURE: Ambient **STORAGE PRESSURE:** Ambient

GENERAL: Store in a cool, dry, well ventilated area away from all sources of ignition and incompatible materials and products. Protect eyes, skin and clothing from contact with this material. Wear recommended personnel protective equipment. Avoid breathing vapors, mists or aerosols. Use with adequate ventilation. Do not take internally. Wash thoroughly after handling this material.

The empty containers may be hazardous. They may contain organic residues that can be ignited and will burn. Do not cut, puncture or weld on or near these containers.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

CONTROL MEASURES: A local or general mechanical exhaust ventilation system is recommended if use generates aerosols, mists or vapors that may cause irritation.

RECOMMENDED PERSONAL PROTECTIVE EQUIPMENT

RESPIRATOR: Respiratory protection is not normally required. If use creates vapors, mists or aerosols that may cause irritation, then a NIOSH approved full facepiece or half mask cartridge respirator with a good particulate / mist cartridge or supplied air is recommended.

EYES: Wear chemical goggles (recommended by ANSI Z87.1-1979), unless a full facepiece respirator is worn.

GLOVES: Neoprene, polyvinyl chloride (PVC) or butyl rubber gloves.

CLOTHING & EQUIPMENT: Wear a neoprene or PVC apron when handling this material. An eye wash station and safety shower should be available in the work area.

FOOTWEAR: Neoprene or butyl rubber boots.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance:	Clear, colorless	**Bulk Density (pounds/ft^3):**	Not applicable
Physical State:	Liquid	**Vapor Pressure:**	About 0.1 mm Hg @ 20 °C
Odor:	Very slight, mild	**Vapor Density (air=1):**	2.62
Odor Threshold:	No data available	**Evaporation Rate (n-Butyl Acetate=1):**	No data available
Molecular Formula:	$Ch_3CH(OH)CH_2OH$	**VOC Content:**	Approximately 1,038 g/L
Molecular Weight:	76.10	**% Volatile:**	100
Boiling Point:	Approximately 187.4 °C (369 °F)	**Solubility in H_2O:**	Complete
Freezing/Melting Point:	Approximately -60 °C (-76 °F)	**Octanol/Water Partition Coefficient:**	Log P_{oct}: -1.41
Specific Gravity:	Approximately 1.038 @ 20 °C	**pH (as is):**	Not applicable
Density (pounds/gallon):	Approximately 8.66	**pH (1% solution):**	No data available

10. STABILITY AND REACTIVITY

GENERAL: This product is stable and hazardous polymerization will not occur.

CONDITIONS TO AVOID: None known.

INCOMPATIBLE MATERIAL: Strong oxidizers and materials reactive with hydroxyl compounds.

HAZARDOUS DECOMPOSITION PRODUCTS: When heated to decomposition, it emits toxic oxides of carbon plus possible dense smoke.

SENSITIVITY TO MECHANICAL IMPACT: This material is not sensitive to mechanical impact.

SENSITIVITY TO STATIC DISCHARGE: This material is not sensitive to normal static discharge.

11. TOXICOLOGICAL INFORMATION

Components:	**Propylene Glycol**
Eye Contact:	Rabbit: 500 mg/24 hours; Mild
Skin Contact:	Human: 500 mg/7 days; Mild
Oral Rat LD_{50}:	20 gm/kg
Dermal Rabbit LD_{50}:	20,800 mg/kg
Inhalation Rat LC_{50}:	No data available
Human Data:	Oral Child TDLo: 79 gm/kg/56 weeks-I; Changes in surface EEG, Convulsions
Other Toxicological Data:	Oral Mouse LD_{50}: 22 gm/kg
Carcinogenicity:	No data available
Teratogenicity:	Intraperitoneal Mouse TDLo: 100 mg/kg (15 Days pregnant) Fetotoxicity
Mutagenicity:	DNA Inhibition - Mouse Subcutaneous: 8,000 mg/kg
Synergistic Products:	None Reported
Target Organs:	Eyes, Skin
Medical Conditions Aggravated By Exposure:	Skin Disorders

12. ECOLOGICAL INFORMATION

ENVIRONMENTAL FATE:

The environmental fate of this material is expected to be: **Land:** biodegradation with some leaching into the groundwater. **Water:** biodegradation. **Air:** expected to exist in vapor phase only due to low vapor pressure. This material is not expected to bioaccumulate.

ENVIRONMENTAL CONSIDERATIONS:

The aquatic toxicity for Propylene Glycol is: Gold fish LC_{50} (24 hours): Greater than 5,000 mg/L. Guppies LC_{50} (48 hours): Greater than 10,000 mg/L. Fingerling Rainbow trout LC_{50} (24 hours): Greater than 50,000 mg/L.

13. DISPOSAL CONSIDERATIONS

RCRA 40 CFR 261 CLASSIFICATON: Non-Hazardous Waste

U.S. EPA WASTE NUMBER/DESCRIPTION: Not applicable

If this product is disposed of as shipped, it does not meet the criteria of a hazardous waste as defined under 40 CFR 261, in that it does not exhibit the characteristics of a hazardous waste of Subpart C, nor is it listed as a hazardous waste under Subpart D due to toxicity. As a non-hazardous liquid waste, it should be disposed of in accordance with all local, state, and federal regulations. Consult state or local officials for proper disposal method.

14. TRANSPORTATION INFORMATION

DOT PROPER SHIPPING NAME: **Not Regulated**

Hazard Class: N/A **UN Number:** N/A **Packing Group:** N/A

Primary Label: None Required **Subsidiary Label(s):** None Required

Primary/Subsidiary Placards: None Required

DOT Reportable Quantity (RQ): None

Marine Pollutant: No

DOT 2000 North American Emergency Response Guidebook No.: **None**

TDG PROPER SHIPPING NAME: **Not Regulated**

Hazard Class: N/A **UN Number:** N/A **Packing Group:** N/A

Primary Label: None Required **Subsidiary Label(s):** None Required

Primary/Subsidiary Placards: None Required

TDG Reportable Quantity (RQ): # None

TDG Schedule XII: No

Regulated Limit (RL): ## None

Other Shipping Information: None

Canadian Transportation of Dangerous Goods Regulations (TDGR), Part IX, Table I, Quantities or levels for Immediate Reporting: releases of reportable quantities, RQ, that meet the definition of a "dangerous occurrence" (a threat to life, health, property, or the environment) must be reported to the appropriate authorities as outlined in TDGR 9.13(1) and 9.14(1).

Reporting to Environment Canada is required for any releases exceeding the regulated limits, RL, of 9.2 materials (primary or secondary). The regulated limits are found in Schedule XII of the TDGR.

15. REGULATORY INFORMATION

OMPONENTS: Propylene Glycol

OSHA Target Organs: Eyes, Skin

Carcinogenic Potential:

Regulated by OSHA:	No
Listed on NTP Report:	No
Listed by IARC:	No
IARC Group:	Not applicable
ACGIH Appendix A:	Not listed
A1 Confirmed Human:	Not applicable
A2 Suspected Human:	Not applicable

U.S. EPA Requirements

Release Reporting

CERCLA (40 CFR 302)

Listed Substance:	Not listed
Reportable Quantity:	Not applicable
Category:	Not applicable
RCRA Waste No.:	Not applicable
Unlisted Substance:	Not applicable
Reportable Quantity:	Not applicable
Characteristic:	Not applicable
RCRA Waste No.:	Not applicable

SARA TITLE III

Section 302 & 303 (40 CFR 355):

Listed Substance:	Not listed
Reportable Quantity:	Not applicable
Planning Threshold:	Not applicable

Section 311 & 312 (40 CFR 370):

Hazard Categories (**product**): **Fire: N** **Sudden Release of Pressure: N** **Reactive: N** **Acute Health: N** **Chronic Health: N**

Planning threshold: 10,000 pounds

Section 313 (40 CFR 372):

Listed Toxic Chemical:	Not listed
Reporting Threshold:	Not applicable

U.S. TSCA Status

Listed (40 CFR 710): Yes

State Regulations

State of California: Safe Drinking Water and Toxins Enforcement Act, 1986 (Proposition 65):

Carcinogen:	No
Reproductive Toxin:	No

Other Regulations

State Right To Know Laws: PA

Canadian Regulations

Product Information:

Controlled Product:	**No**
WHMIS Hazard Symbols:	**Not applicable**
WHMIS Class & Division:	**Not applicable**

Ingredient Information:

IDL Substance:	Yes
Domestic Substance List:	Yes

16. OTHER INFORMATION

EPA Registration number: Not applicable

Approved Product Uses: Not applicable

Special Notes:

None

MSDS Revision Information: Information Revised This Issue Date: **Revised per the Canadian 3 year update requirement.**

Form Revision made 08/01/00

MSDS Distributed by: GW International

Environmental Department

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Prepared By:	Edward Doheny	Date Prepared:	June 14, 2000

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